Mathematics Revision Test 30

1. Write **70 + 6 +** $\frac{9}{100}$ as a **Decimal**: _____

2. Find in **Lowest Terms**: $\frac{8}{2}$ = _____ $\frac{25}{100}$ = _____

3. Add 5 to 7 squared: _____

4. From the sum of 3.6 and 6.2 take 5: _____

5. Ring the fractions which are equivalent to $\frac{3}{4}$:

$\frac{6}{10}$	$\frac{9}{12}$	$\frac{4}{5}$	$\frac{15}{20}$

6. 3 hrs 40 minutes:_____ minutes

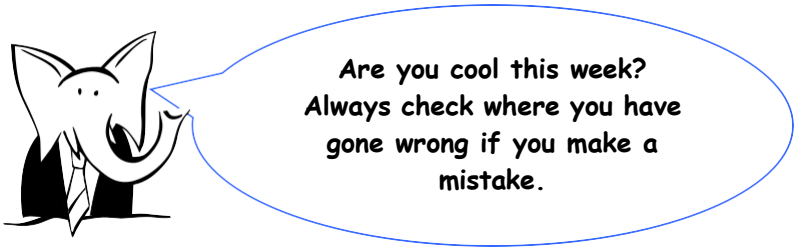
7. Complete the next two numbers in the series:
   0.01 0.1 1 _____ _____

8. 0.5 litres = 330ml +_____ ml

9.

   Angle x = _____ Angle y _____

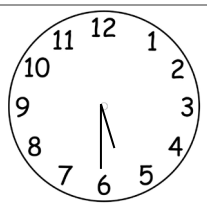
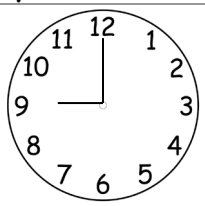
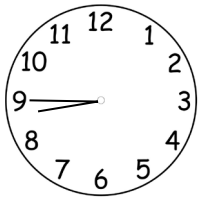
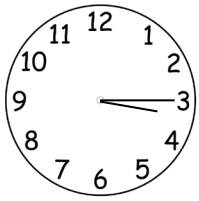
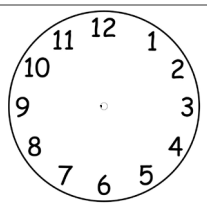
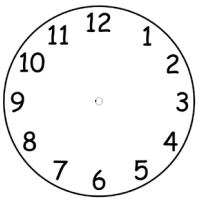
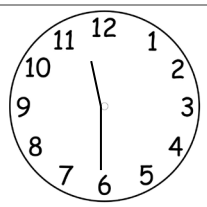
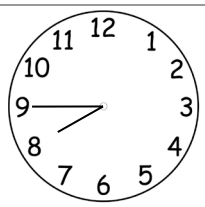
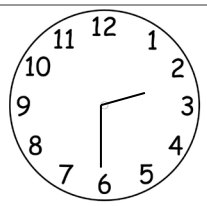
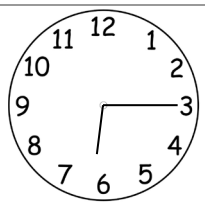
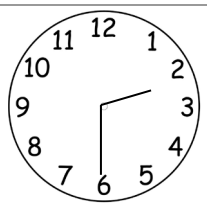
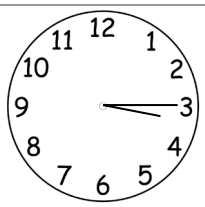
10. How many times is 260.6 greater than 2.606? _____



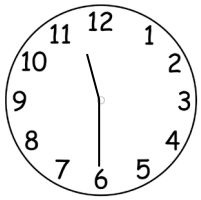
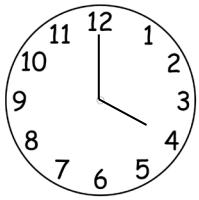
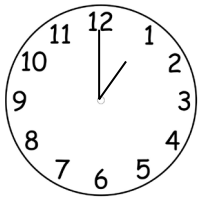
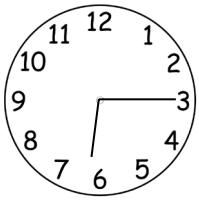
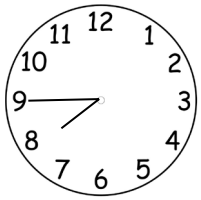
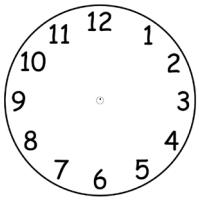
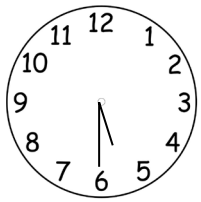
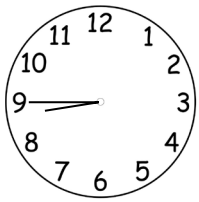
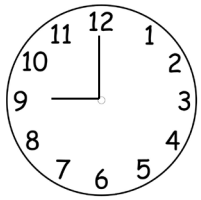
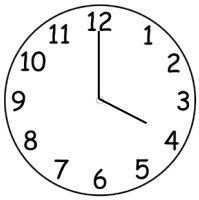
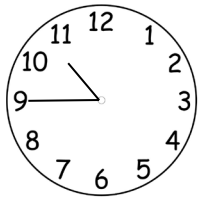
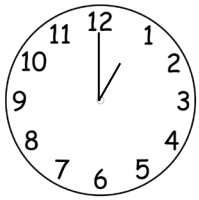
MARKS:.....

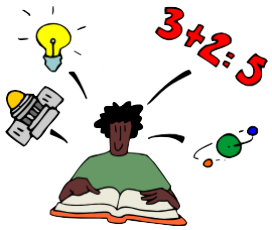
24 Hour Clock

Convert these times into the 24 hour clock. The first one has been done for you

 am	05.30	 am	
 am		 pm	
 am		 midnight	
 pm		 pm	
 pm		 pm	
 am		 am	

24 Hour Clock

 am		 pm	
 am		 am	
 am		 noon	
 pm		 pm	
 pm		 am	
 pm		 pm	



Calculation Ladder

Speed Test!

Ladder 8

1.	9
2.	n^2
3.	eighteen more
4.	$66\frac{2}{3}$ % of this
5.	find 50%
6.	÷ 11
7.	Multiply by 12
8.	$\sqrt{n}$
9.	$33\frac{1}{3}$ % of this
10.	Multiply by 9
	Your answer is

How many seconds this week? _____

1. Answer the following calculation questions.

a. 632
× 8
= _____

b. 723
386
+ 714
= _____

c. 917
- 48
= _____

d. 345
× 7
= _____

e. 685
720
+ 635
= _____

f. 410
- 28
= _____

g. 562
- 71
= _____

h. 387
× 6
= _____

i. 435
456
+ 895
= _____

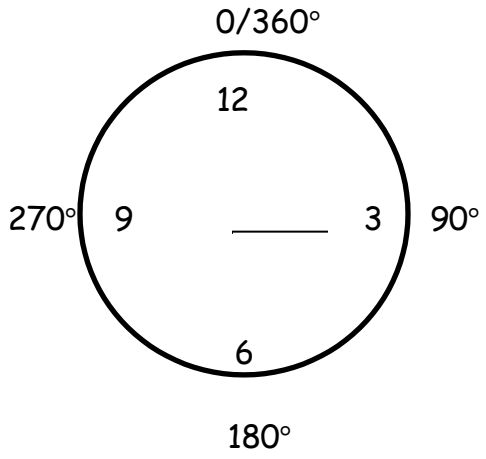
j. 546
- 39
= _____

k. 424
× 7
= _____

l. 725
236
+ 546
= _____

---

2. Consider the diagram and then answer the questions.



This is a clock face showing the time is 3 o'clock.
A clock is a circle and we have marked degrees around the edge of it.
If the minute hand moved from pointing to 12 to pointing to 3 it will have moved through 90°.

a. How many degrees does the hour hand move through between 12 o'clock and 4 o'clock? _____

b. I set out for work at 7.15, I got there at 7.55. through how many degrees had the minute hand moved during that time?_____

c. What angle is represented by the movement of the minute hand between 10.30 and 10.45? _____

d. I started watching the football match at 3.45 and finished at 5.00. through how many degrees had the minute hand moved during that time?

_____

e. What angle is represented if the minute hand moves from its position at 7 o'clock to its position at 7.07 and 30 seconds? _____

3. Work out the missing digits:

a.

```
  3 5 6
+ 2 __ __

= __ 7 1
```

b.

```
  __ 9 __
+ 4 2 3

= 7 __ 7
```

c.

```
  __ 9 6
+ 1 __ 3

= 4 3 __
```

d.

```
  __ 7 9
- 2 __ __

= 3 6 3
```

---

4. Find the highest common factor of the following numbers.

Example: The highest common factor of 12 and 6 is 6.

a. 12 and 8 _____

b. 15 and 6 _____

c. 36 and 22 _____

---

5. Answer the following percentage questions.

a. What is 30% of £120? _____

b. Find 25% of 200cm? _____

c. Find 75% 1.2kg? _____

d. If I spend 36% of my money what percentage of my money is left?

_____

e. If I started with £3 and spent 40% what would be left? _____

f. If one angle in a triangle is 20° and another angle in the triangle is 30% larger, how many degrees is the larger angle? _____

6. Answer the following questions on the area of triangles.

To find the area of a triangle we multiply it's base by its height and divide by two. Work out the area of the following triangles expressing your answer in sq cm.

a. Triangle base 5cm, triangle height 8cm. _____

b. Triangle base 6cm, triangle height 7cm. _____

c. Triangle base 8cm, triangle height 7cm. _____

---

7. Answer the following questions on the area of rectangles.

With a rectangle the area is worked out by multiplying the length by the breadth. Work out the area of the following rectangles, expressing your answer in suitable units.

a. Length 27cm, breadth 18cm. _____

b. Length 7 miles, breadth 3 miles. _____

c. Length 16mm, breadth 4mm. _____

d. Length 8km, breadth 6km. _____

---

8. Answer the following worded division and multiplication questions.

a. A machine knits 36 pairs of socks per hour, how many individual socks will it make in 7 hours? _____

b. A bus driver makes £6.56 per hour after tax. If he works for 8 hours per day and five days per week how much money will he make?_____

c. A packet of sweets holds 29 sweets. Three girls want to share the packet equally. How many sweets will each girl get? _____

d. How many would be left over? _____

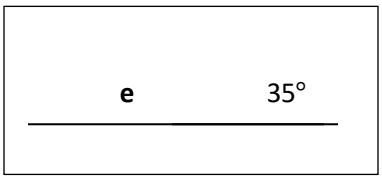
9. Calculate the missing angles in each of the questions and diagrams below.

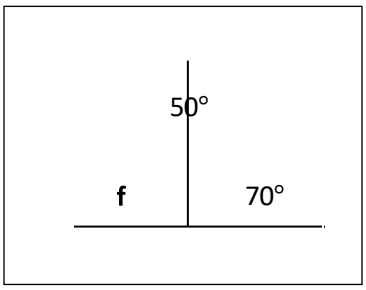
a. What do three angles within every triangle add up to? _____

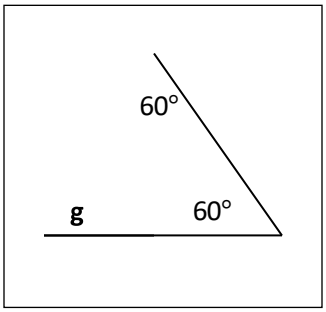
b. What do the angles on a straight line add up to? _____

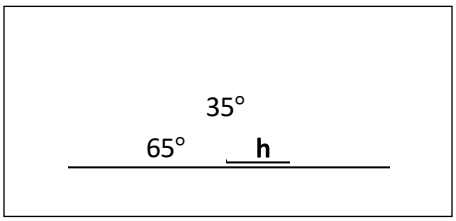
c. An equilateral triangle has three angles. What size is each angle? _____

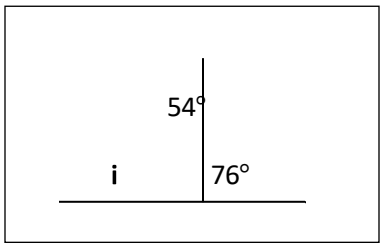
d. What do the angles around a point add up to? _____

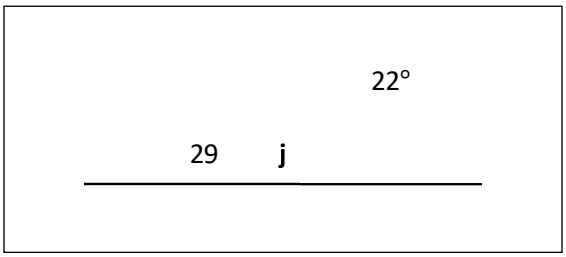


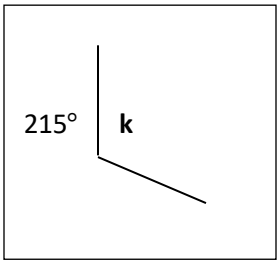


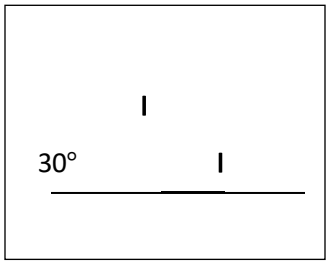










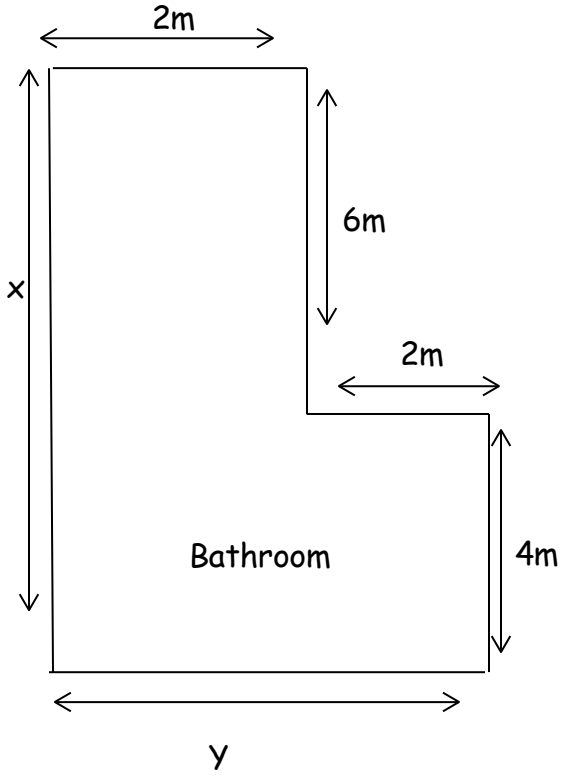


e = _____ **f** = _____ **g** = _____ **h** = _____

i = _____ **j**= _____ **k** = _____ **l**= _____

4

10. Study this diagram of a bathroom and then answer the questions below.



a. If a bathroom measured 12ft by 12ft what would its area be? _____

b. If a decorative strip of wallpaper 1 metre high was put on top of the wall around the whole edge of the bathroom which is illustrated, what area of wall would be covered by wallpaper? _____

c. What length of the wallpaper would be needed? _____

d. What is the length of the bathroom wall marked x? _____

e. What is the width of the bathroom wall marked y? _____

f. What is the total area of the bathroom floor?

---

11. Answer the following worded questions about money and weight.

a. If I buy four steaks for £3.26 each how much change will I receive if I pay with a £20 note? _____

b. Mike pays for himself and four friends to take the bus. Each ticket costs £1.25. How much does Mike pay in total? _____

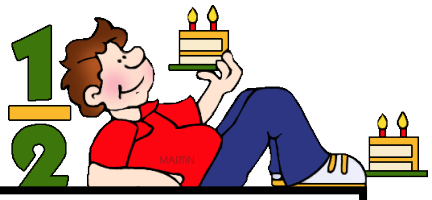
c. My mother wanted to weigh a large piece of meat. She knew it would weigh somewhere between 4kg and 5kg. She had no weights for her scales and so planned to use a number of cans of baked beans (300g each) and she also had three small packets of nuts which weighed 100g each. If the meat weighed 4.7kg how many cans of beans and packets of nuts would she need to get closest to the true weight of the meat? _____

END OF TEST

If you have time

- **Go back and complete any questions you missed out.**
- **Check your answers carefully.**

Maths Reasoning



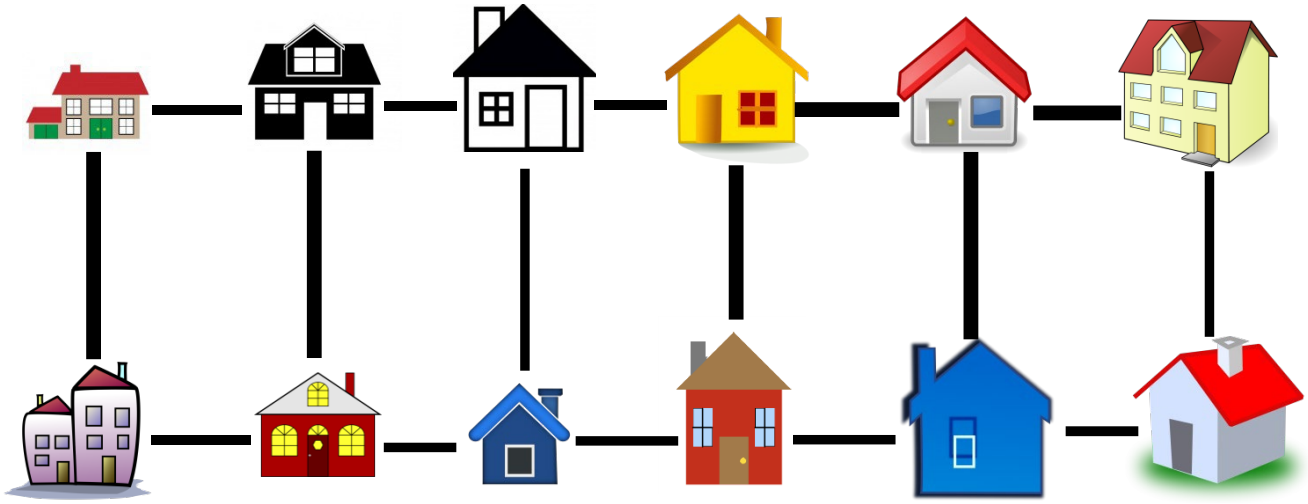
These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

In Townville there are 2 rows of 6 houses that are linked by 16 paths.

Each path is 20m long.

Gillian is tired but needs to deliver a letter to every house.

Her route can start and end at any house, but she must go along each path at least once.



How long is the shortest route that Gillian can take?

What is the shortest route if there are 4 rows of 5 houses?

Maths

Revision Sheet

1. 76.09

2. $\frac{2}{5}$ $\frac{1}{4}$

3. 54

4. 4.8

5 $\frac{9}{12}$ and $\frac{15}{2}$

1. 220 mins

2. 10, 100

3. 170 mls

4. **x** = 82° **y** = 98°

- see note below

5. 100 times

Note for question 9:
You will need to remember that there are 180° in a straight line. X must be 82° (opposite angles are equal) so 180° - 82° will give you the answer for y.

24 Hour Clock

Sheet 1

05.30	09.00
08.45	15.15
10.45	00.00
23.30	19.45
14.30	18.15
02.30	03.15

Sheet 2

11.30	16.00
01.00	06.15
07.45	12.00
17.30	20.45
21.00	04.00
22.45	13.00

Calculation Ladder

Answer = 18

ANSWERS - COMPLETE MATHS – PART 32

Maths
Type 1
Paper 7

Page 1

1.
a. 5056
b. 1823
c. 869
d. 2415
e. 2040
f. 382
g. 491
h. 2322
i. 1786
j. 507
k. 2968
l. 1507

1.
a. 120°
b. 240°
c. right angle/90°
d. 450°

Page 2

e. 45°

3.
a.
3 5 6
+ 2 1 5
= 5 7 1
b.
2 9 4
+ 4 2 3
= 7 1 7
c.
2 9 6
+ 1 4 3
= 4 3 9
d.
5 7 9
- 2 1 6
= 3 6 3

4.
18. 4
19. 3
20. 2

5.
a. £36
b. 50cm
c. 0.9kg
d. 64%
e. £1.80
f. 26°

Page 3

6.
a. 20cm²
b. 21cm²
c. 28cm²

7.
a. 486cm²
b. 21 sq mile
c. 64 sq mm
d. 48 sq km

8.
a. 504
b. £262.40
c. 9
d. 2

Page 4

9.
a. 180°
b. 180
c. 60
d. 360°
e. 145°
f. 60°
g. 60°
h. 80°
i. 50°
j. 129°
k. 145°
l. 75°

Page 5

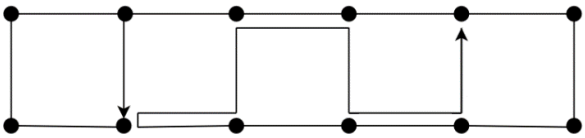
10.
a. 144 sq ft
b. 28 m²
c. 28m
d. 10m
e. 4m
f. 28m²

11.
a. £6.96
b. £6.25
c. 15 cans of beans & 2 packets of nuts

Maths Reasoning 21

(There are lots of routes that will get the same answer so don't worry if you don't get the same route as below)

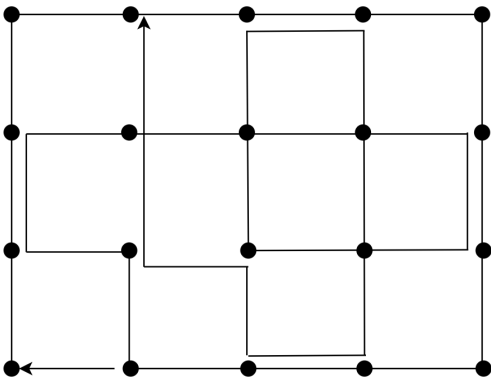
With 2 rows of 6 houses, the shortest route takes Gillian along 19 paths.



So, the shortest route is:

$$Shortest\ route = 19 * 20\ metres = 360\ metres$$

With 4 rows of 5 houses, the shortest route takes Gillian along 35 paths.



So, the shortest route is:

$$Shortest\ route = 35 * 20\ metres = 700\ metres$$